

Initiating Search

November 10, 2025, 10:03 PM

• Search:

CAS Lexicon Search:

Bucherer-Bergs reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (73)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (35)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

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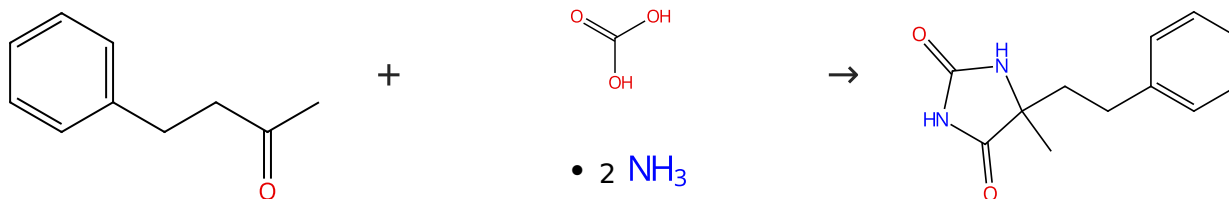
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Reactions (35)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (77)

 Suppliers (90)

 Suppliers (34)

31-173-CAS-21384928

Steps: 1 Yield: 100%

Direct chemical derivatization of natural plant extract: straight forward synthesis of natural plant-like hydantoin

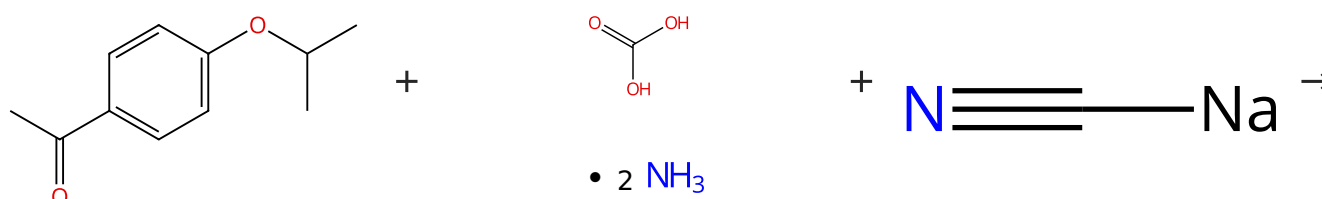
By: Tomohara, Keisuke; et al

Tetrahedron Letters (2016), 57(8), 924-927.

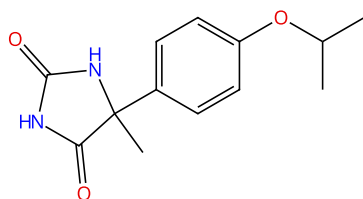
1.1 Reagents: Potassium cyanide
Solvents: Ethanol, Water; 24 h, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%


 Suppliers (58)

 Suppliers (90)

 Suppliers (45)

 Suppliers (15)

31-173-CAS-18122706

Steps: 1 Yield: 99%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β-selective agonists

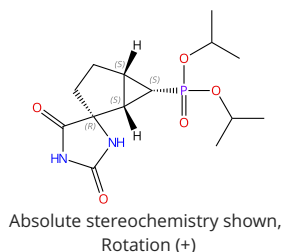
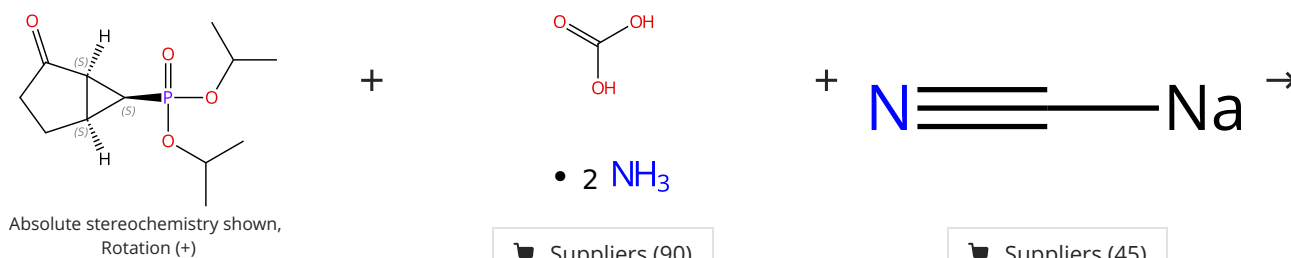
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; rt; 1 h, 100 °C

Scheme 3 (1 Reaction)

Steps: 1 Yield: 98%



31-173-CAS-18114409

Steps: 1 Yield: 98%

Constrained cycloalkyl analogues of glutamic acid: stereoccontrolled synthesis of (+)-2-aminobicyclo[3.1.0]hexane-2,6-dicarboxylic acid (LY354740) and its 6-phosphonic acid analogue

By: Krysiak, Jerzy; et al

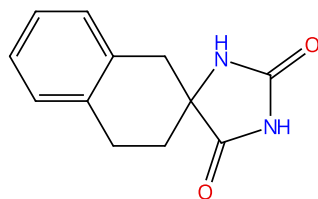
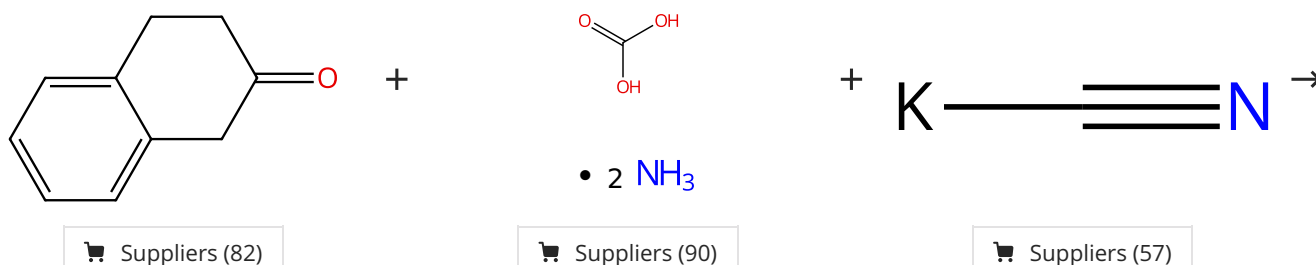
Tetrahedron: Asymmetry (2010), 21(11-12), 1486-1493.

1.1 Solvents: Ethanol, Water; 3 d, 35 °C

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (34)

31-173-CAS-18077775

Steps: 1 Yield: 96%

Continuous Synthesis of Hydantoins: Intensifying the Bucherer-Bergs Reaction

By: Monteiro, Julia L.; et al

Synlett (2016), 27(1), 83-87.

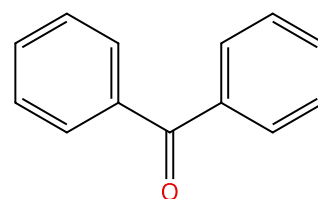
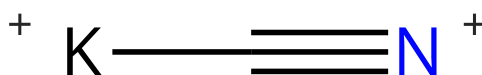
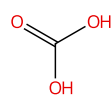
1.1 Solvents: Dimethylformamide, Ethyl acetate, Water; 32 min, 20 bar, 120 °C

1.2 Reagents: Hydrochloric acid
Solvents: Water; acidified

Experimental Protocols

Scheme 5 (2 Reactions)

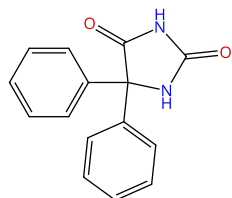
Steps: 1 Yield: 84-90%



Suppliers (90)

Suppliers (57)

Suppliers (136)



Suppliers (108)

31-173-CAS-21075927

Steps: 1 Yield: 90%

Highly efficient and rapid synthesis of diverse hydantoin derivatives using nano-ordered ZnO catalyst under mechanical ball milling

By: Maddah, Bozorgmehr

Iranian Chemical Communication (2017), 5(1), 58-66.

1.1 Catalysts: Zinc oxide (ZnO); 50 min, rt

1.2 Reagents: Hydrochloric acid

Solvents: Water; neutralized

Experimental Protocols

31-173-CAS-18168493

Steps: 1 Yield: 84%

Microwave-promoted facile and rapid synthesis procedure for the efficient synthesis of 5,5-disubstituted hydantoins

By: Safari, Javad; et al

Synthetic Communications (2013), 43(23), 3115-3120.

1.1 Solvents: Ethanol, Water; 10 min; cooled

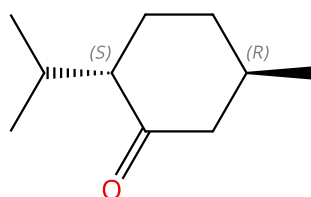
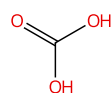
1.2 Reagents: Hydrochloric acid

Solvents: Water; neutralized, cooled

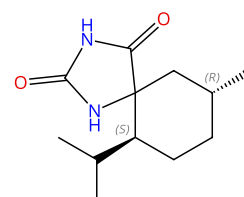
Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 90%



Relative stereochemistry shown



Relative stereochemistry shown



Suppliers (90)

Suppliers (45)

31-173-CAS-21386059

Steps: 1 Yield: 90%

Direct chemical derivatization of natural plant extract: straight forward synthesis of natural plant-like hydantoin

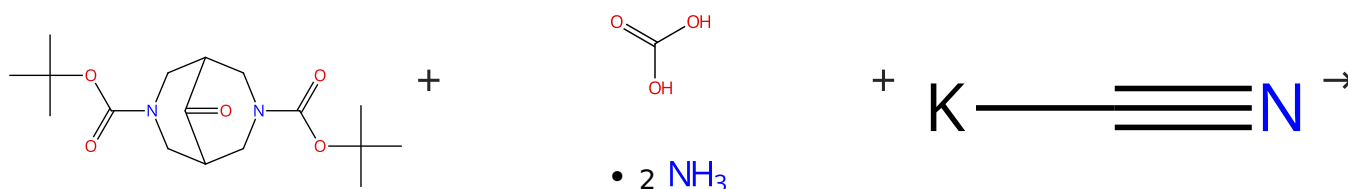
By: Tomohara, Keisuke; et al

Tetrahedron Letters (2016), 57(8), 924-927.

1.1 Reagents: Potassium cyanide
Solvents: Ethanol, Water; 24 h, rt

Scheme 7 (1 Reaction)

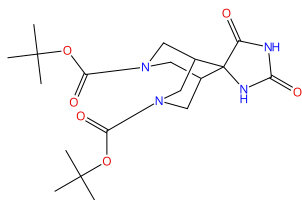
Steps: 1 Yield: 88%



Suppliers (22)

Suppliers (90)

Suppliers (57)



31-173-CAS-18027885

Steps: 1 Yield: 88%

Synthesis of a bis-cationic α,α -disubstituted amino acid (9-amino-bispidine-9-carboxylic acid) and its effects on the conformational properties of peptides

1.1 Solvents: Ethanol, Water; 15 h, 120 °C

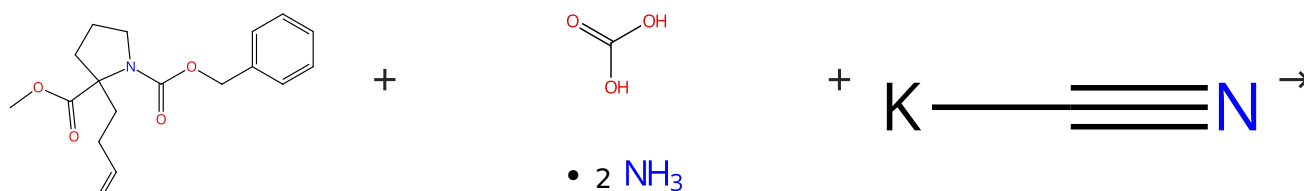
Experimental Protocols

By: Yamashita, Hiroko; et al

Tetrahedron (2015), 71(15), 2241-2245.

Scheme 8 (1 Reaction)

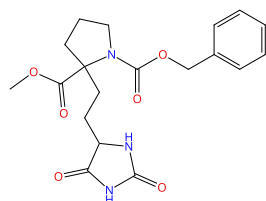
Steps: 1 Yield: 86%



Supplier (1)

Suppliers (90)

Suppliers (57)



31-173-CAS-18060912

Steps: 1 Yield: 86%

Synthesis and evaluation of C8-substituted 4.5-spiro lactams as Glycogen Phosphorylase a inhibitors

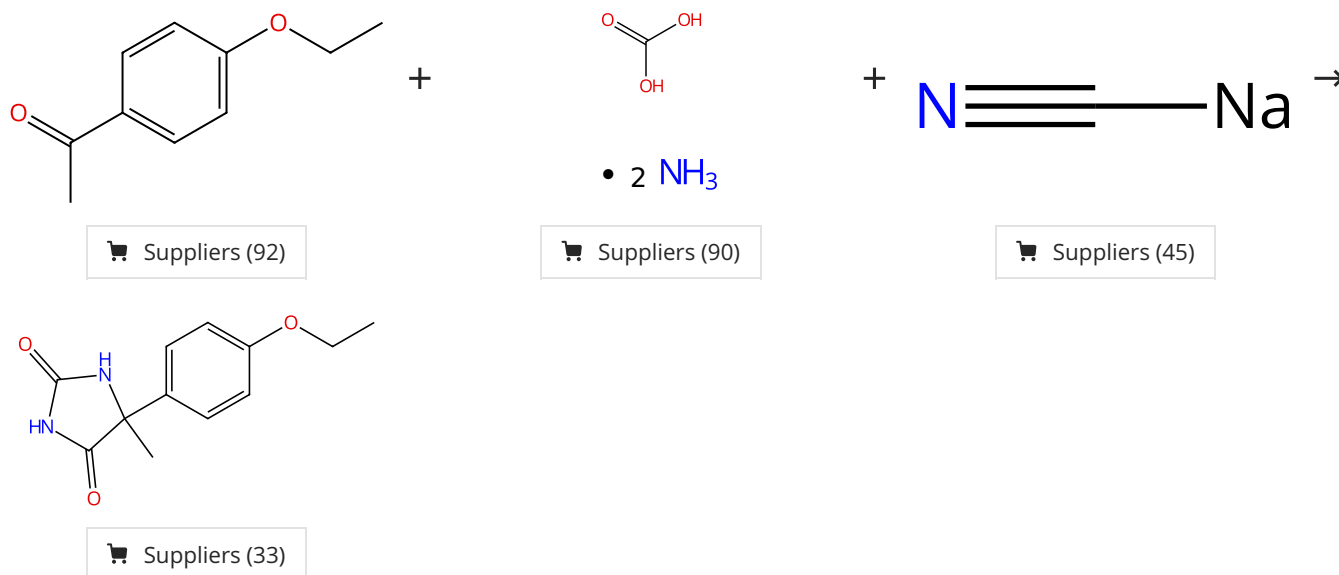
1.1 Solvents: Methanol, Water; 10 min, rt; 84 h, rt

By: Loughlin, Wendy A.; et al

Tetrahedron (2013), 69(5), 1576-1582.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 85%



31-173-CAS-18125986

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

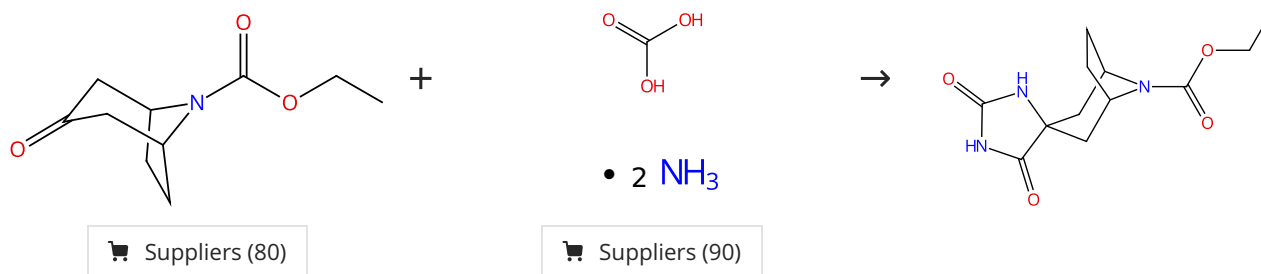
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 10 (1 Reaction)

Steps: 1 Yield: 85%



31-173-CAS-21386459

Steps: 1 Yield: 85%

Direct chemical derivatization of natural plant extract: straight forward synthesis of natural plant-like hydantoin

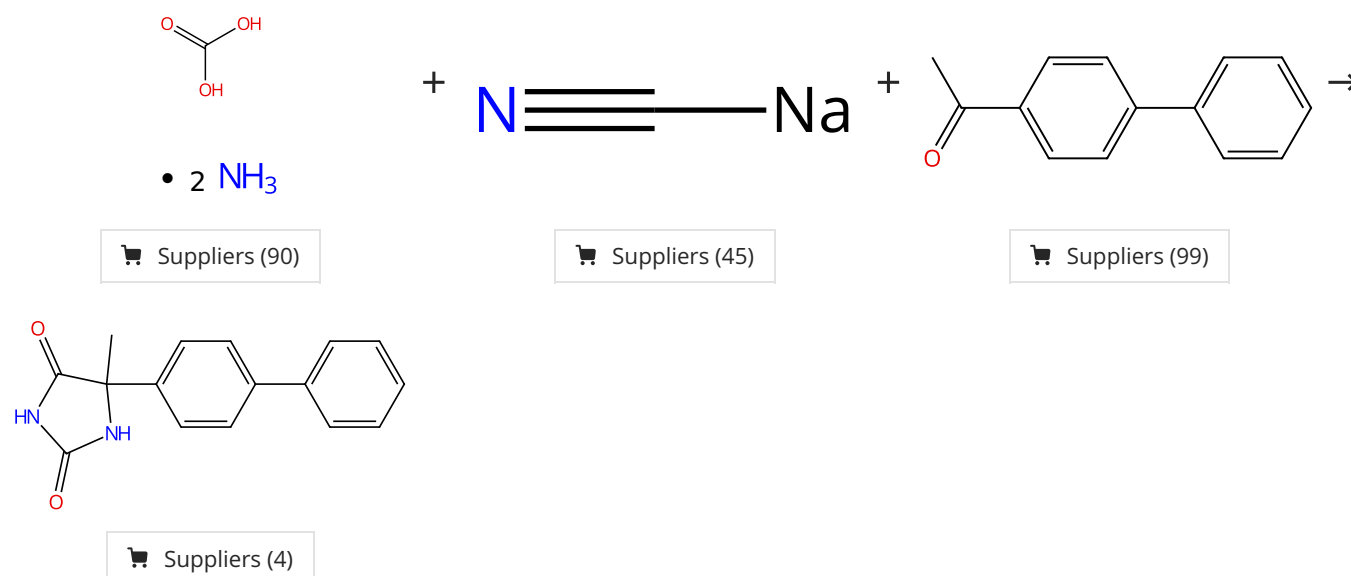
By: Tomohara, Keisuke; et al

Tetrahedron Letters (2016), 57(8), 924-927.

1.1 Reagents: Potassium cyanide
Solvents: Ethanol, Water; 24 h, rt

Scheme 11 (1 Reaction)

Steps: 1 Yield: 85%



31-173-CAS-18192153

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

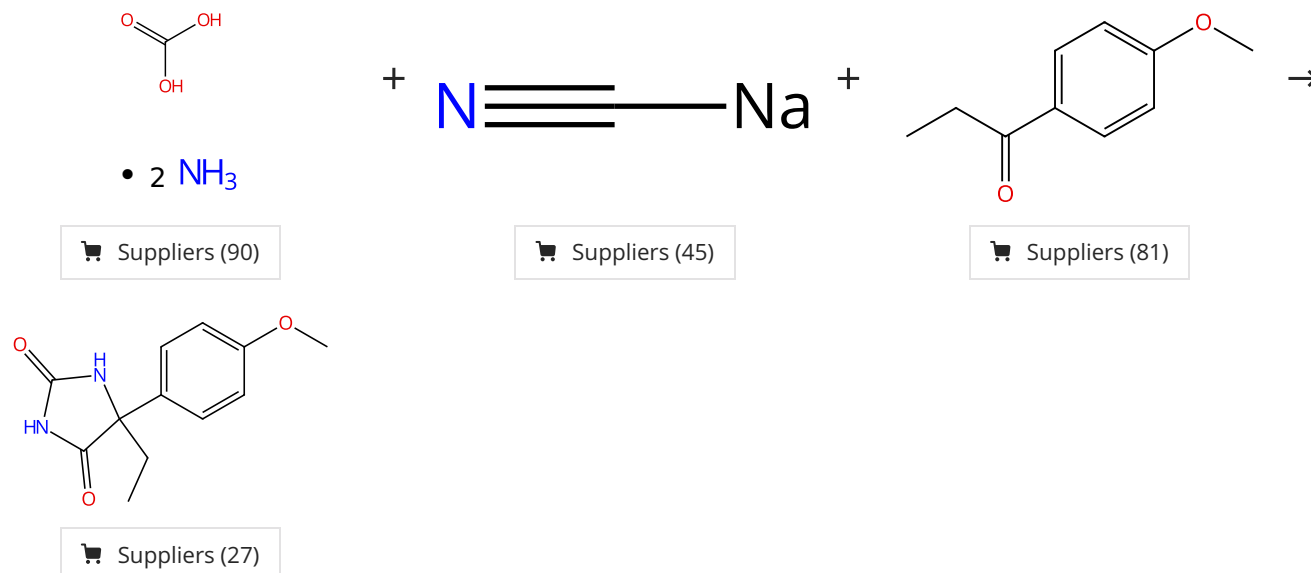
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 12 (1 Reaction)

Steps: 1 Yield: 85%



31-173-CAS-18185635

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

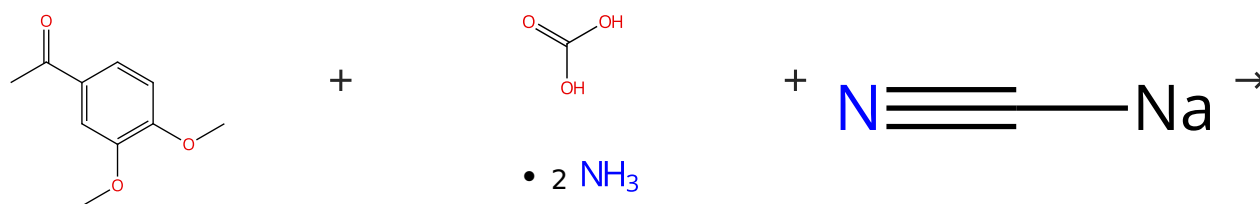
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 13 (1 Reaction)

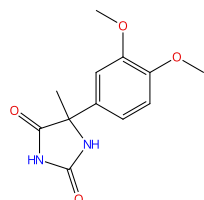
Steps: 1 Yield: 85%



Suppliers (83)

Suppliers (90)

Suppliers (45)



Suppliers (21)

31-173-CAS-18120135

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

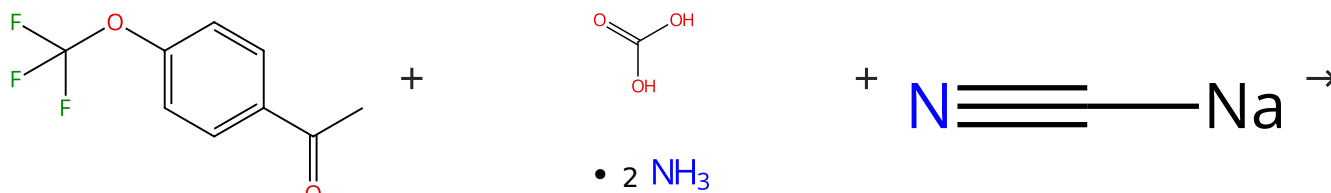
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 14 (1 Reaction)

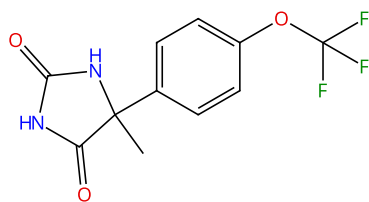
Steps: 1 Yield: 85%



Suppliers (73)

Suppliers (90)

Suppliers (45)



Suppliers (29)

31-173-CAS-18128373

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

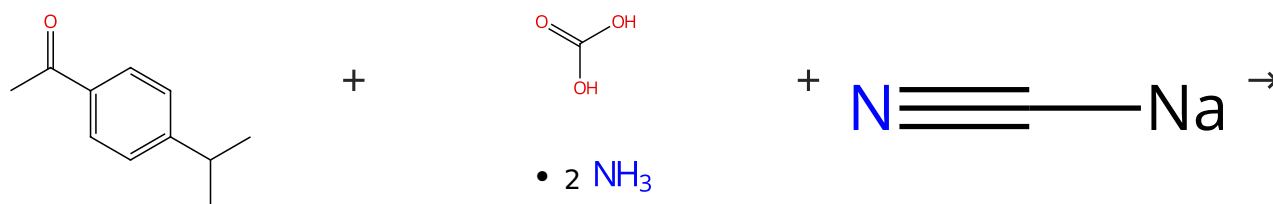
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 15 (1 Reaction)

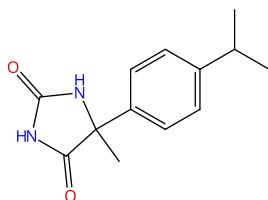
Steps: 1 Yield: 85%



Suppliers (81)

Suppliers (90)

Suppliers (45)



Suppliers (27)

31-173-CAS-18117726

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

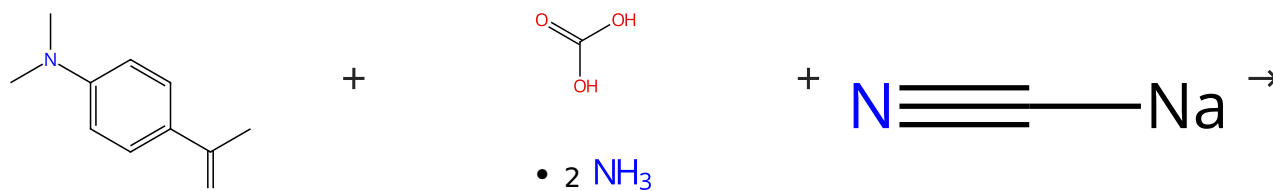
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 16 (1 Reaction)

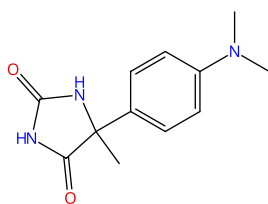
Steps: 1 Yield: 85%



Suppliers (77)

Suppliers (90)

Suppliers (45)



Suppliers (2)

31-173-CAS-18134043

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

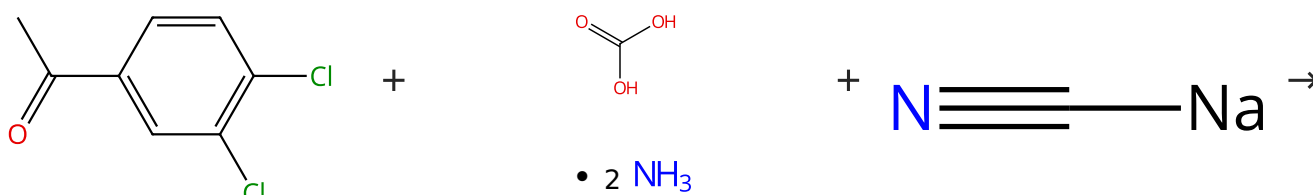
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 17 (1 Reaction)

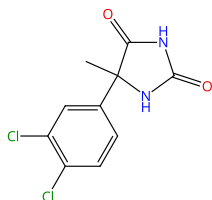
Steps: 1 Yield: 85%



Suppliers (80)

Suppliers (90)

Suppliers (45)



Suppliers (33)

31-173-CAS-18194042

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

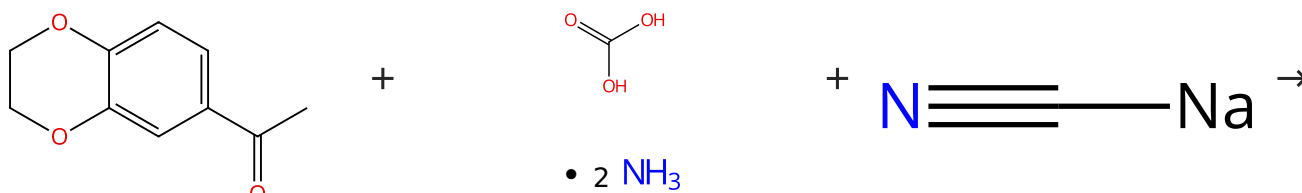
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 18 (1 Reaction)

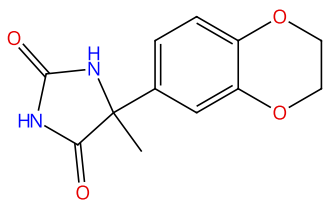
Steps: 1 Yield: 85%



Suppliers (90)

Suppliers (90)

Suppliers (45)



Suppliers (29)

31-173-CAS-18132740

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

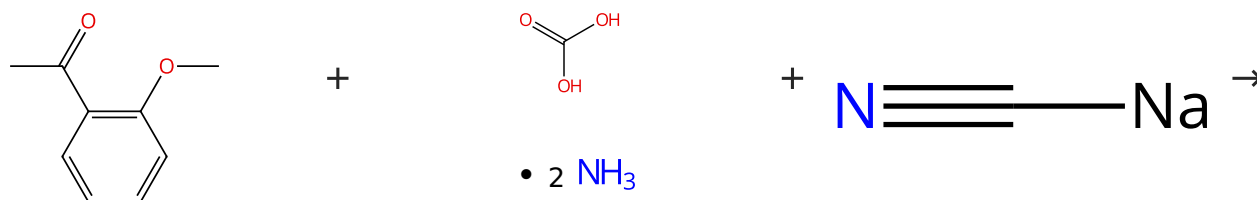
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 19 (1 Reaction)

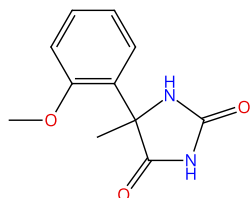
Steps: 1 Yield: 85%



Suppliers (87)

Suppliers (90)

Suppliers (45)



Suppliers (30)

31-173-CAS-18115089

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

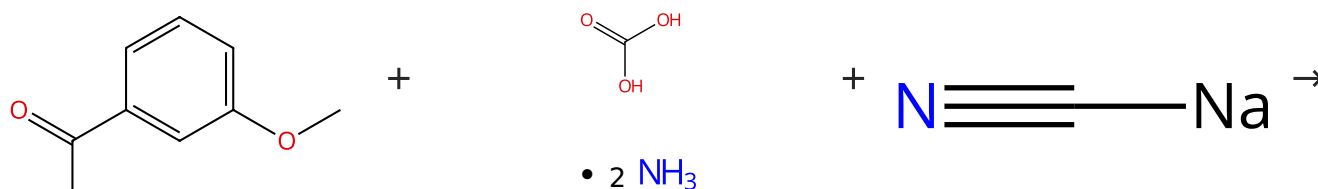
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 20 (1 Reaction)

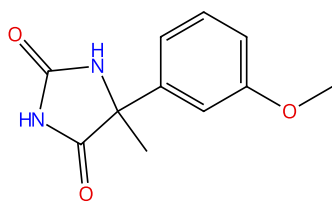
Steps: 1 Yield: 85%



Suppliers (90)

Suppliers (90)

Suppliers (45)



Suppliers (33)

31-173-CAS-18121989

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

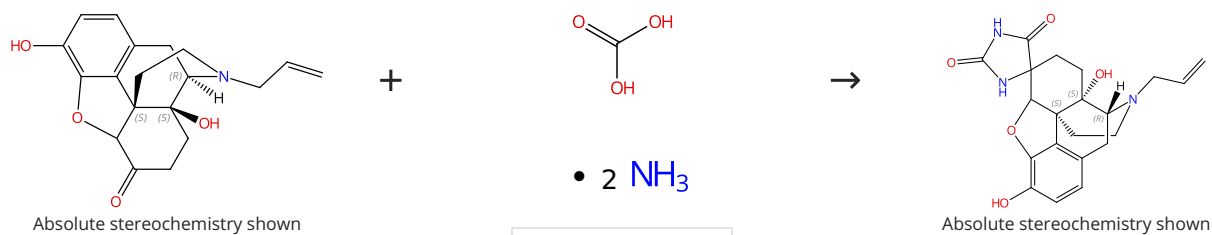
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 21 (1 Reaction)

Steps: 1 Yield: 85%



Supplier (1)

Suppliers (90)

31-173-CAS-21385701

Steps: 1 Yield: 85%

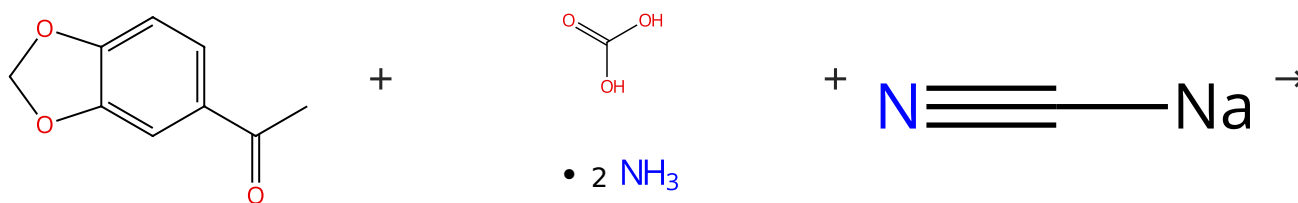
Direct chemical derivatization of natural plant extract: straight forward synthesis of natural plant-like hydantoin

1.1 **Reagents:** Potassium cyanide
Solvents: Ethanol, Water; 24 h, rt

By: Tomohara, Keisuke; et al
 Tetrahedron Letters (2016), 57(8), 924-927.

Scheme 22 (1 Reaction)

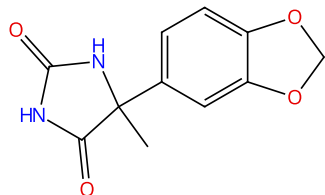
Steps: 1 Yield: 85%



Suppliers (75)

Suppliers (90)

Suppliers (45)



Suppliers (36)

31-173-CAS-18130331

Steps: 1 Yield: 85%

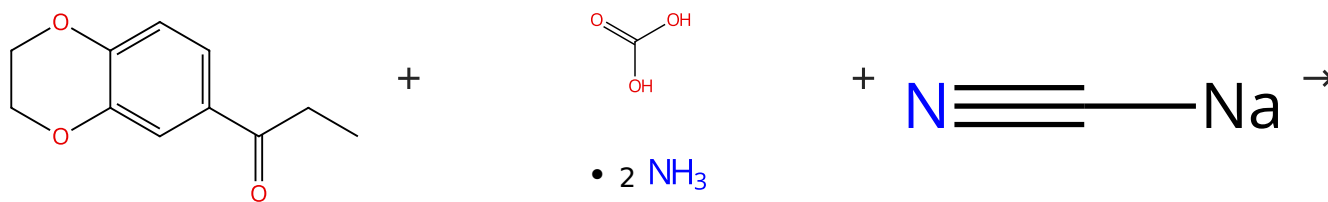
Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

1.1 **Solvents:** Ethanol, Water; 48 h, 100 °C

By: Koura, Minoru; et al
 Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

Scheme 23 (1 Reaction)

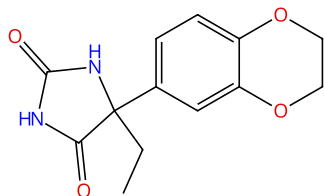
Steps: 1 Yield: 85%



Suppliers (59)

Suppliers (90)

Suppliers (45)



Suppliers (3)

31-173-CAS-18180499

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

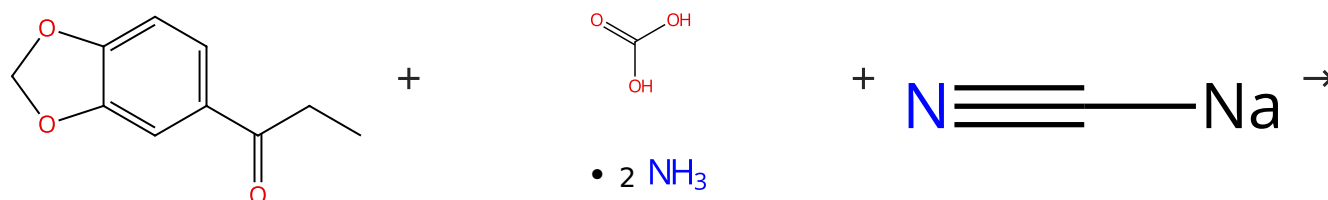
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 24 (1 Reaction)

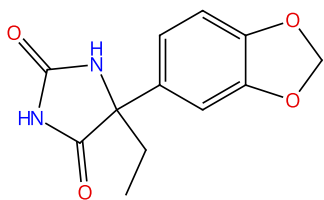
Steps: 1 Yield: 85%



Suppliers (51)

Suppliers (90)

Suppliers (45)



Suppliers (4)

31-173-CAS-18184834

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

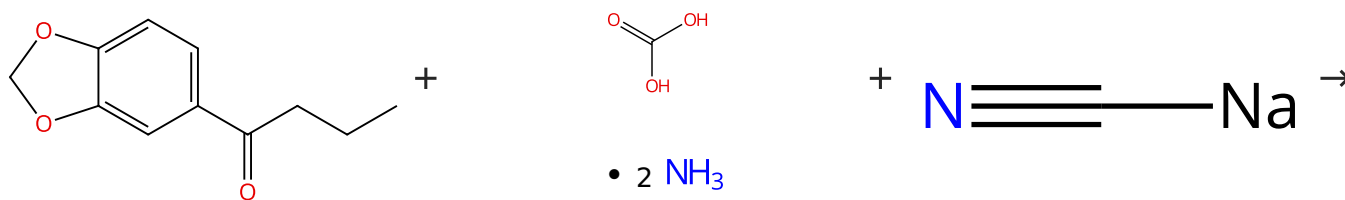
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 25 (1 Reaction)

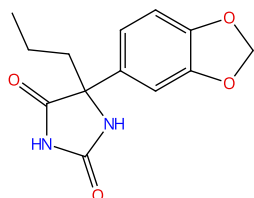
Steps: 1 Yield: 85%



Suppliers (61)

Suppliers (90)

Suppliers (45)



31-173-CAS-18182897

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

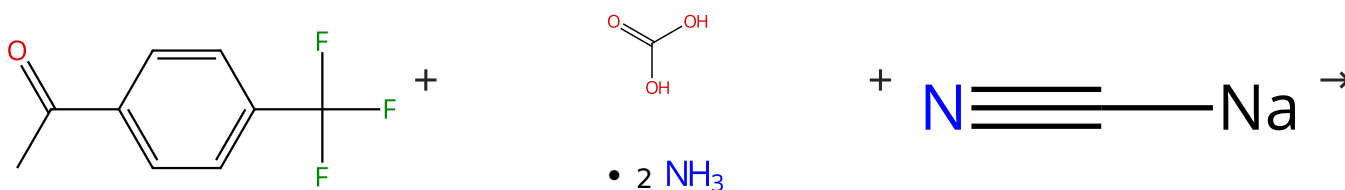
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 26 (1 Reaction)

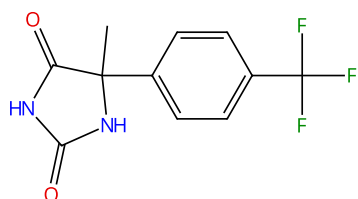
Steps: 1 Yield: 85%



Suppliers (86)

Suppliers (90)

Suppliers (45)



Suppliers (31)

31-173-CAS-18190278

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

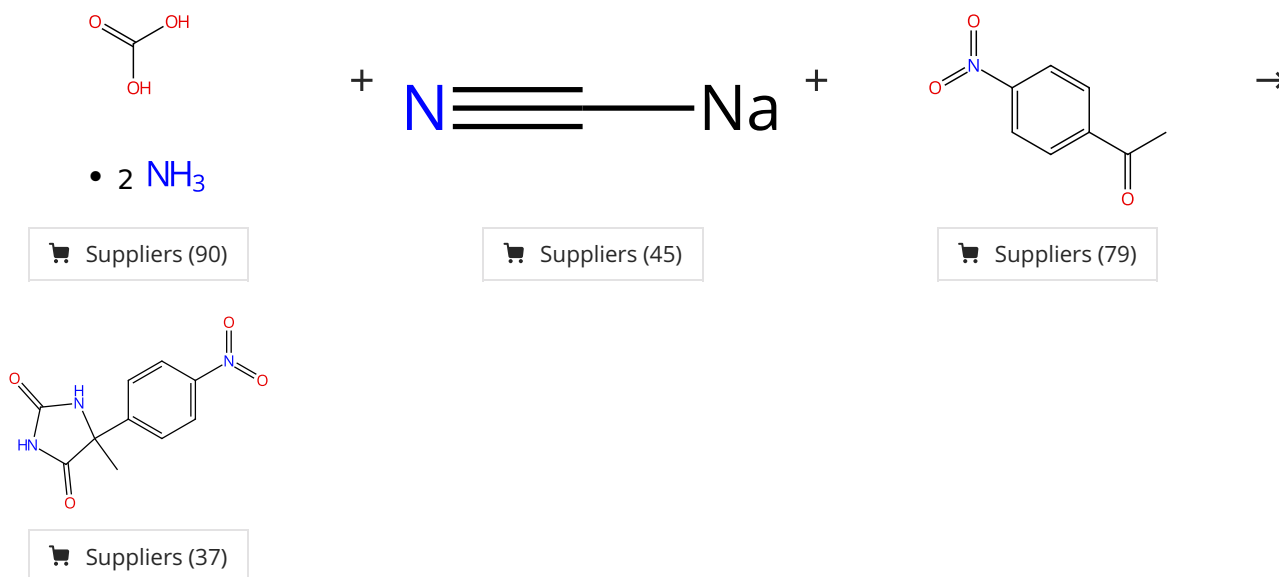
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 27 (1 Reaction)

Steps: 1 Yield: 85%



31-173-CAS-18187953

Steps: 1 Yield: 85%

Design, synthesis and pharmacology of 1,1-bistrifluoromethyl carbinol derivatives as liver X receptor β -selective agonists

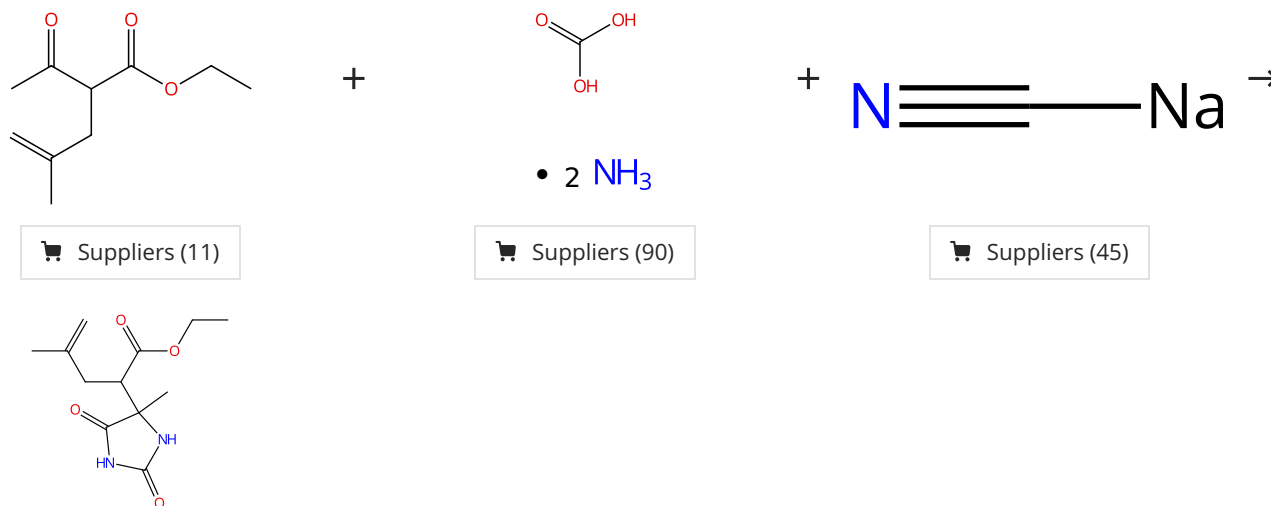
By: Koura, Minoru; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(13), 2668-2674.

1.1 Solvents: Ethanol, Water; 48 h, 100 °C

Scheme 28 (1 Reaction)

Steps: 1 Yield: 83%



31-173-CAS-18147393

Steps: 1 Yield: 83%

Three-step synthetic pathway to fused bicyclic hydantoin derivatives involving a selenocyclization step

By: Smit, Biljana M.; et al

Tetrahedron (2015), 71(7), 1101-1108.

1.1 Solvents: Ethanol, Water; 24 h, rt $\rightarrow$ reflux; reflux $\rightarrow$ rt

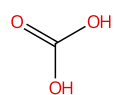
1.2 Reagents: Hydrochloric acid

Solvents: Water; pH 2, rt; overnight, cooled

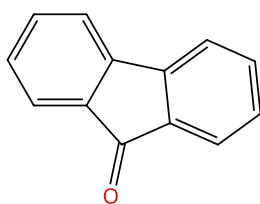
Experimental Protocols

Scheme 29 (1 Reaction)

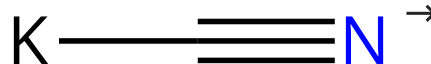
Steps: 1 Yield: 83%



+



+



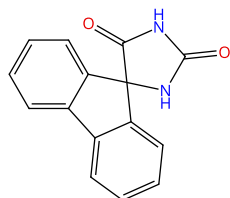
→



Suppliers (90)

Suppliers (100)

Suppliers (57)



Suppliers (24)

31-173-CAS-18053291

Steps: 1 Yield: 83%

Microwave-promoted facile and rapid synthesis procedure for the efficient synthesis of 5,5-disubstituted hydantoins

By: Safari, Javad; et al

Synthetic Communications (2013), 43(23), 3115-3120.

1.1 Solvents: Ethanol, Water; 13 min; cooled

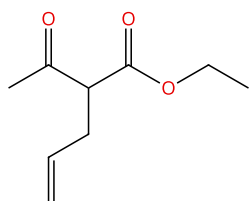
1.2 Reagents: Hydrochloric acid

Solvents: Water; neutralized, cooled

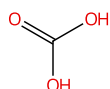
Experimental Protocols

Scheme 30 (1 Reaction)

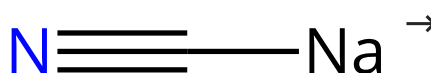
Steps: 1 Yield: 83%



+



+



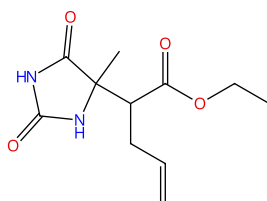
→



Suppliers (50)

Suppliers (90)

Suppliers (45)



31-173-CAS-18153787

Steps: 1 Yield: 83%

Three-step synthetic pathway to fused bicyclic hydantoins involving a selenocyclization step

By: Smit, Biljana M.; et al

Tetrahedron (2015), 71(7), 1101-1108.

1.1 Solvents: Ethanol, Water; 24 h, rt → reflux; reflux → rt

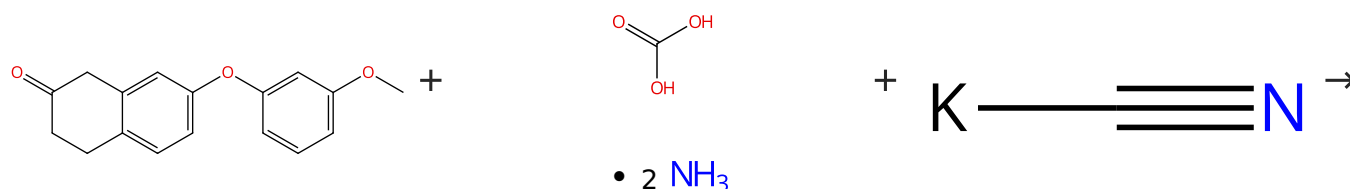
1.2 Reagents: Hydrochloric acid

Solvents: Water; pH 2, rt; overnight, cooled

Experimental Protocols

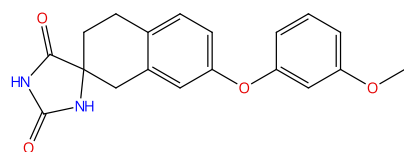
Scheme 31 (1 Reaction)

Steps: 1 Yield: 79%



Suppliers (90)

Suppliers (57)



31-173-CAS-21306715

Steps: 1 Yield: 79%

Synthesis and antitussive effect of new hydantoin compounds

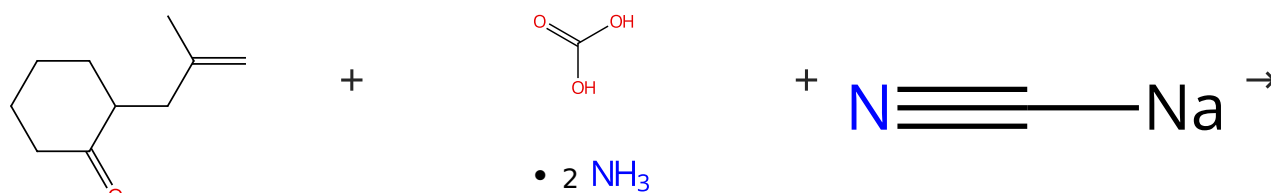
By: Zhang, Qun-li; et al

Chemical Research in Chinese Universities (2013), 29(1), 76-81.

1.1 Solvents: Ethanol, Water; 2 h, reflux

Scheme 32 (1 Reaction)

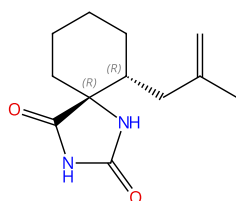
Steps: 1 Yield: 77%



Suppliers (7)

Suppliers (90)

Suppliers (45)



Relative stereochemistry shown

31-173-CAS-21343224

Steps: 1 Yield: 77%

Synthesis of angularly fused (homo)triquinane-type hydantoins as precursors of bicyclic prolines

By: Smit, Biljana; et al

Synthesis (2016), 48(3), 387-393.

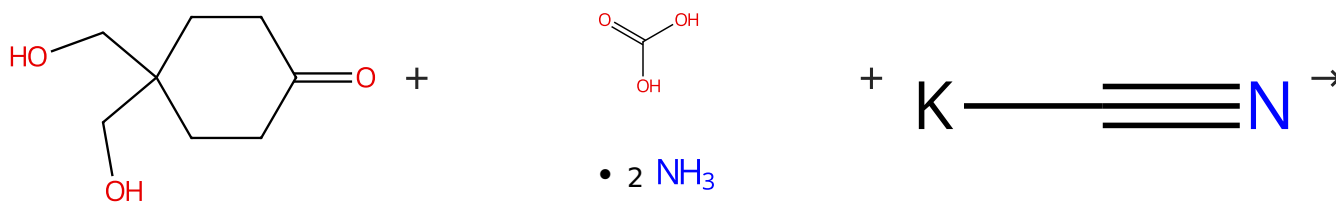
1.1 Solvents: Ethanol, Water; 24 h, rt → reflux; reflux → rt

1.2 Reagents: Hydrochloric acid
Solvents: Water; pH 2

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 72%



Suppliers (27)

Suppliers (90)

Suppliers (57)

31-173-CAS-18039931

Steps: 1 Yield: 72%

Model synthesis of six-membered carbocyclic spironucleosides

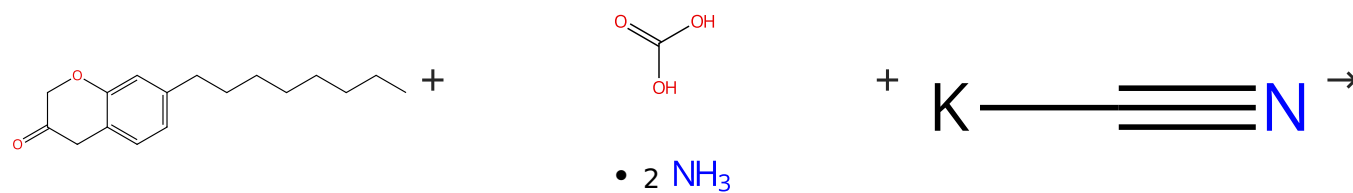
1.1 **Reagents:** Ammonium chloride
Solvents: Ethanol, Water; 16 h, 65 °C

By: Nencka, Radim; et al

Collection of Czechoslovak Chemical Communications (2010), 75(12), 1259-1272.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 70%



Suppliers (90)

Suppliers (57)

31-173-CAS-21379440

Steps: 1 Yield: 70%

Synthesis and immunosuppressive activity of amino alcohol containing coumarin and benzothiophene1.1 **Solvents:** Ethanol, Water; 2 h, reflux

By: Wu, Yi; et al

Experimental Protocols

Chemical Research in Chinese Universities (2016), 32(3), 357-365.